INCIDENT INVESTIGATION REPORT

Security Operations Center (SOC) | Internal Confidential

1. Executive Summary

On **March 14, 2024**, at **5:23PM**, a high-severity security incident was identified on workstation **"Tony" (172.16.17.206)**. The incident involves the download and execution of a sophisticated, multi-stage **Information Stealer** identified as **payload_1.ps1**. Investigation confirmed this as a **True positive** involving successful Command and Control (C2) communication and data exfiltration.

2. Incident Metadata

Incident ID	INC-2026-238
Report Date	March 14, 2024
Incident Category	Malicious Code Execution – Malware Infection - Spyware
Severity Level	High
Assigned Analyst	Kesha Patel
Status	Contained

3. Target & Attacker Information

Affected Asset	Tony (Workstation)
Target IP Address	172.16.17.206
Attacker Source IP	218.92.0.56
File Name	payload_1.ps1
File Path	C:\Users\LetsDefend\Downloads\payload_1.ps1
File hash	db8be06ba6d2d3595dd0c86654a48cfc4c0c5408fdd3f4e1eaf342ac7a2479d0
Active User	LetsDefend

Associated PID	4315
Malicious Domain(C2)	kionagranada.com
C2 IP Address	161.22.46.148
Exfiltration IP	91.236.116.163

4. Investigation Timeline

05:22 PM: The user LetsDefend used Chrome to download a file named payload_1.ps1 from an external Amazon S3 bucket. The device allowed this download, and the file was saved to the user's Downloads folder.

05:23 PM: The user executed the PowerShell script (PID: 4315). Immediately after starting, PowerShell made a DNS Query (Event ID 22) for the domain kionagranada.com, which resolved to the IP address 161.22.46.148.

05:23 PM: The script established a successful connection to the malicious IP (161.22.46.148). A Script Block (Event ID 4104) was captured showing the script running a command to download and execute a second payload, sd2.ps1, directly into the system's memory.

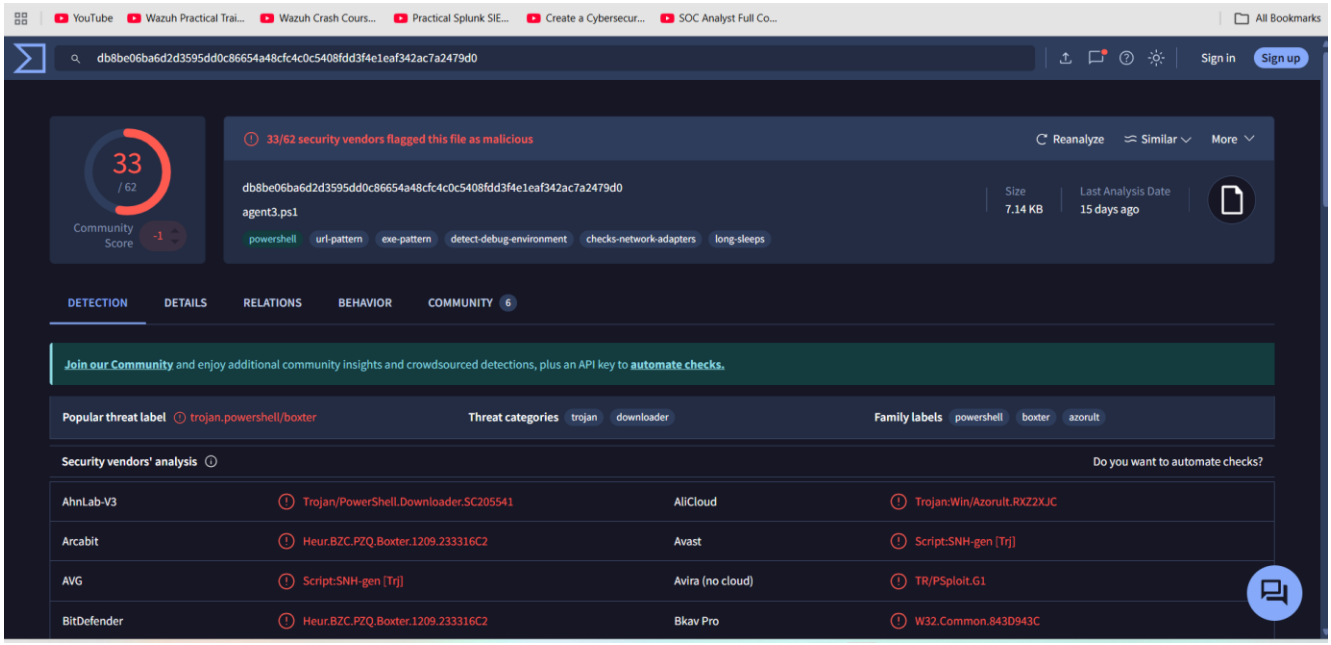
05:23 PM: The malware began communicating with a different external server at 91.236.116.163. A GET request was observed sending a unique system ID and other data to the attacker. The server responded with a 200 OK, confirming the data was successfully received by the attacker.

05:40 PM: Even after the initial execution, the host Tony continued to show active outbound traffic to another suspicious IP address (66.81.199.52). This confirmed that the infection was persistent and the malware was still running.

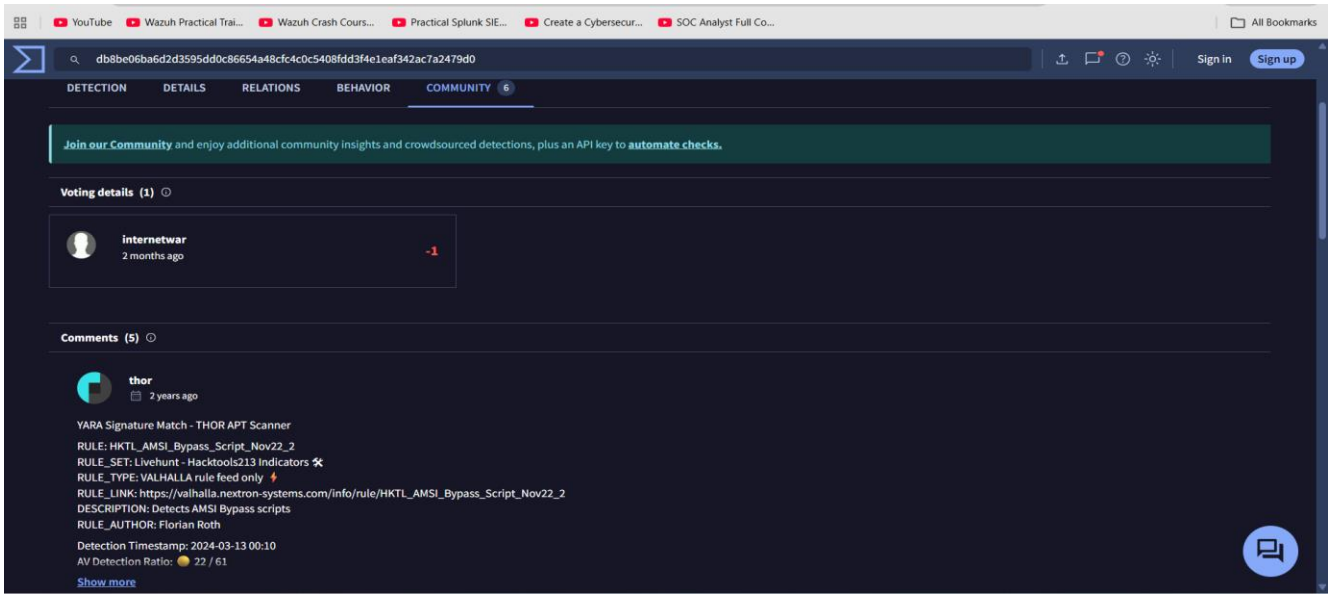
05:45 PM: Following the discovery of active data exfiltration and C2 communication, the host was isolated from the network to stop the attack.

5. Evidence (Artifacts)

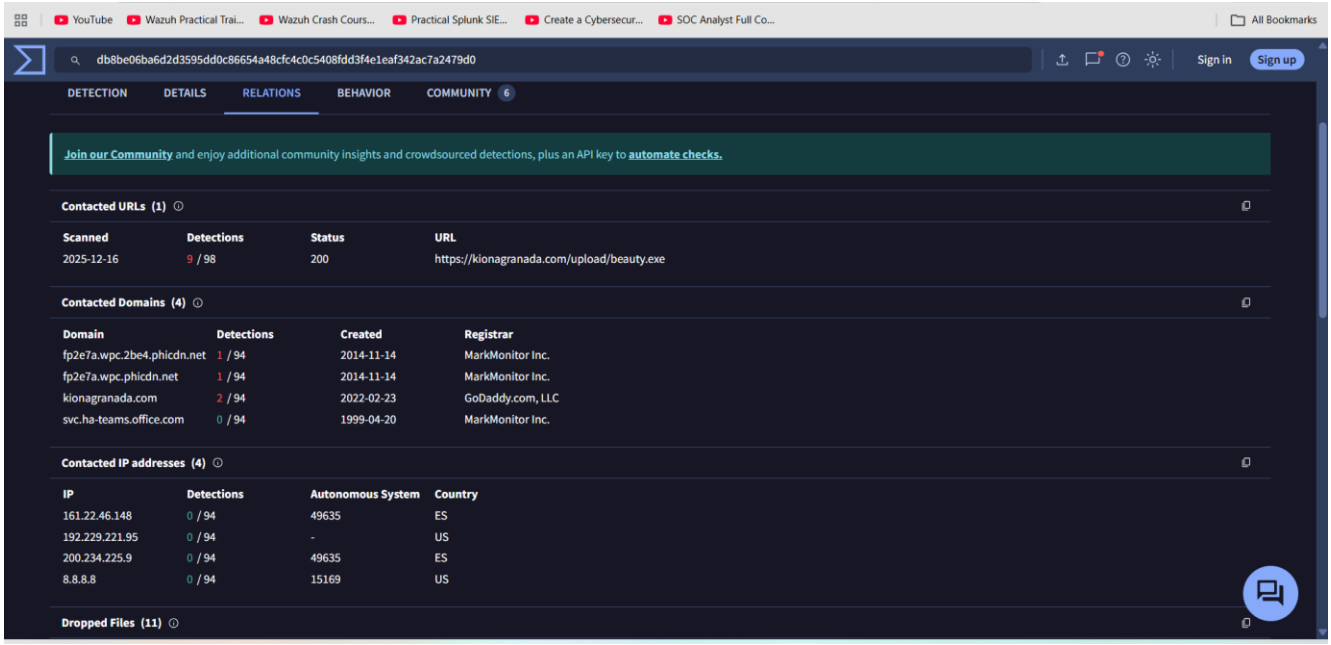
1. Threat Intelligence (Virustotal):

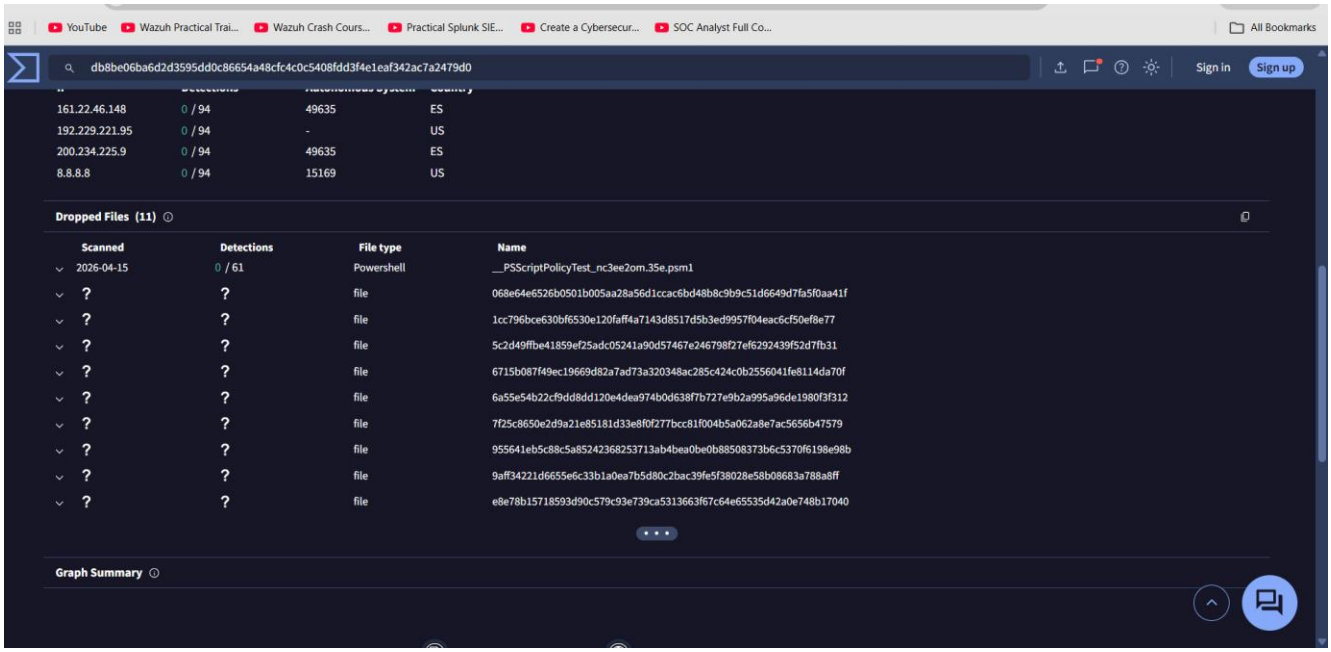


The score shows a 33/62 and community score as -1 which means it is malicious, but we still want to make sure it is so we go in depth.



The Community shows 2 different comments – The THOR APT scanner and the specific YARA signature match.





The relations tab shows the script is connected to a website kionagranada that is hosting virus

Basic properties

MD5	e971b4ed257b0c60287e63c5d533eace
SHA-1	6589114220929c8010b1cc1b8553c0d3e7b3f385
SHA-256	db8be06ba6d2d3595dd0c86654a48cfc4c0c5408fdd3f4e1eaf342ac7a2479d0
Vhash	d235994e52655b7e924429bf3a5e612a
SSDEEP	192:xvOuLf6+qUClbRZqwiJY5AH+W4BJeGZjf6H1XTMTqTn:xvQ+yo+yh
TLSH	T174E179E5BE1C45E419BB226CDF928882AD5DA57415F88505F6BE8C0BF79FE2480F0B0D
File type	Powershell source powershell ps ps1
Magic	ASCII text, with very long lines (4140u), with CRLF line terminators
TrID	file seems to be plain text/ASCII (0%)
Magika	POWERSHELL
File size	7.14 KB (7312 bytes)

History

First Seen In The Wild	2024-03-22 19:52:38 UTC
First Submission	2024-03-12 23:04:41 UTC
Last Submission	2024-03-15 19:05:31 UTC
Last Analysis	2026-03-31 17:07:25 UTC

Names

agent3.ps1
payload_1.ps1

Powershell Info

Cmdlets

YouTube | Wazuh Practical Trai... | Wazuh Crash Cours... | Practical Splunk SIE... | Create a Cybersecur... | SOC Analyst Full Co... | All Bookmarks

db8be06ba6d2d3595dd0c86654a48cfc4c0c5408fdd3f4e1eaf342ac7a2479d0 Sign in Sign up

agent3.ps1
payload_1.ps1

Powershell Info

Cmdlets

new-object
where-object
write-output

Cmdlets Aliases

IWR

.NET Calls

System.MulticastDelegate
System.Reflection.CallingConventions
System.Reflection.Emit.AssemblyBuilderAccess
System.Runtime.InteropServices.HandleRef
System.Runtime.InteropServices.Marshal
reflection.bindingflags

Functions

GDT
GPA

Variables

$null

Details tab – The details tab lists various names like payload_1.ps1 which looks suspicious as hackers use such terms, then the .NET calls have a lot of suspicious names like System. Reflection which is doing something with the memory so this proves the hash is malicious and the incident is a True positive.

YouTube | Wazuh Practical Trai... | Wazuh Crash Cours... | Practical Splunk SIE... | Create a Cybersecur... | SOC Analyst Full Co... | All Bookmarks

LetsDefend NOW PART OF HACKTHEBOX

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Log Management
Case Management
Endpoint Security
Email Security
Threat Intel
Sandbox

Source Address contains "172.16.17.206" All Time

13 events (before Mar, 15, 2024, 11:36 PM UTC)

1 2

Hide Fields

INTERESTING FIELDS
type
source_address
source_port
destination_address
destination_port
raw_log

Event

[Mar, 14, 2024, 05:23 PM] source_address=172.16.17.206 source_port=0 destination_address=172.16.17.206 destination_port=0 raw_log: {"EventID": "4...

[Mar, 14, 2024, 05:23 PM] source_address=172.16.17.206 source_port=49848 destination_address=161.22.46.148 destination_port=443 raw_log: {"Sour...

[Mar, 12, 2024, 07:50 PM] source_address=172.16.17.206 source_port=64436 destination_address=76.13.32.141 destination_port=80 raw_log: {}

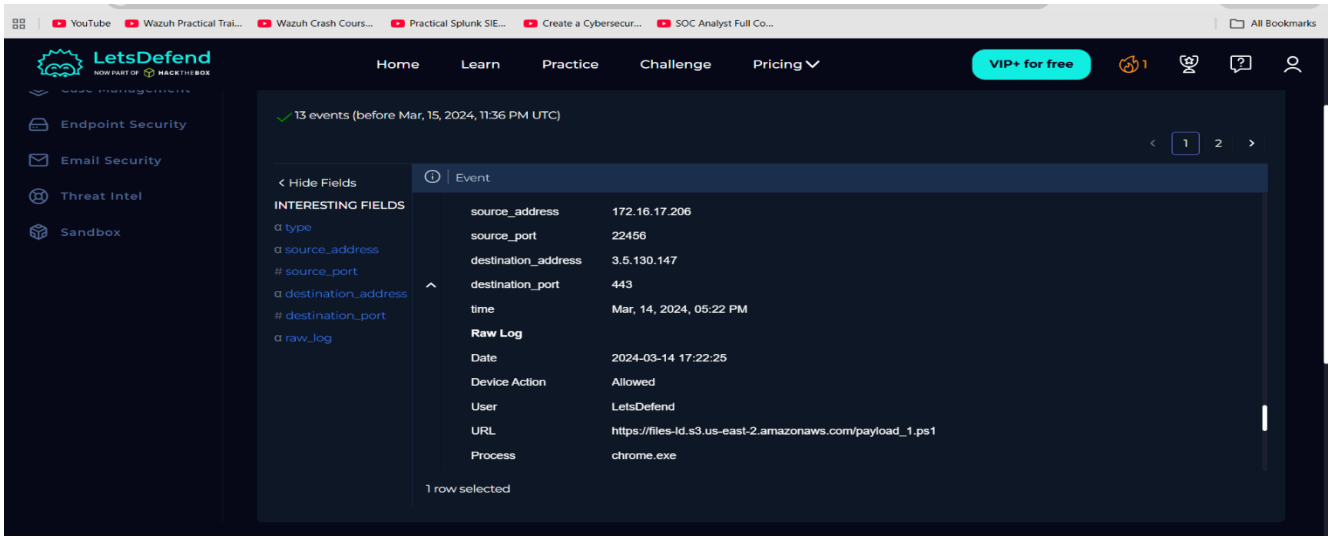
[Mar, 14, 2024, 11:00 PM] source_address=172.16.17.206 source_port=56458 destination_address=157.240.238.174 destination_port=80 raw_log: {}

[Mar, 14, 2024, 05:40 PM] source_address=172.16.17.206 source_port=34771 destination_address=66.81.199.52 destination_port=80 raw_log: {}

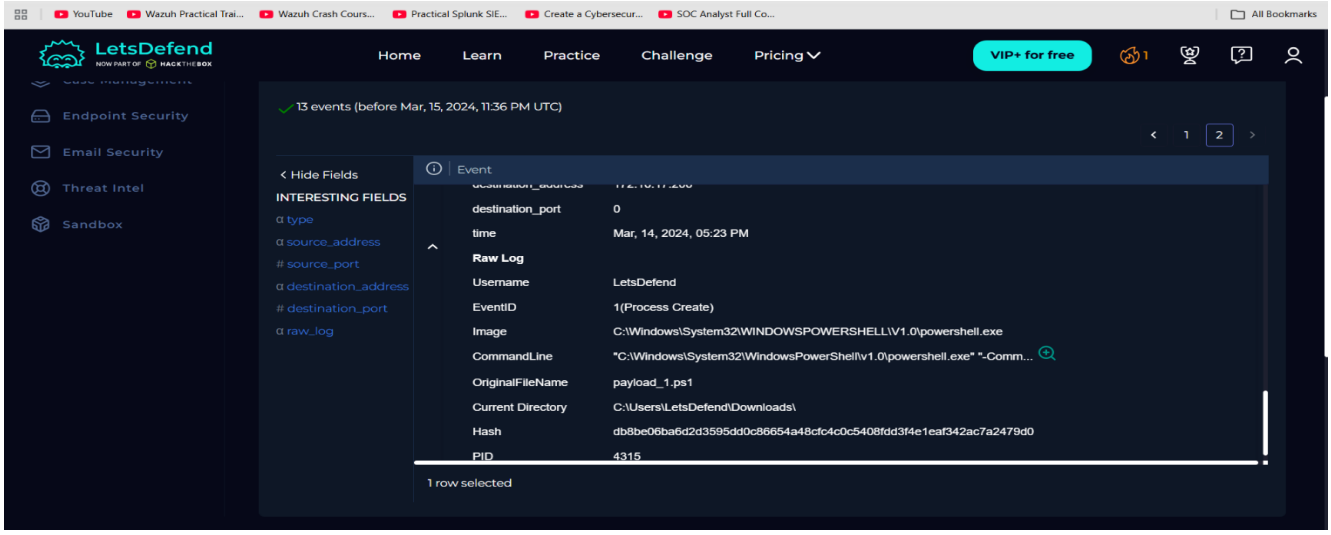
[Mar, 13, 2024, 05:02 PM] source_address=172.16.17.206 source_port=44544 destination_address=151.101.2.217 destination_port=80 raw_log: {}

[Mar, 12, 2024, 06:50 PM] source_address=172.16.17.206 source_port=52978 destination_address=204.79.197.212 destination_port=80 raw_log: {}

Phase 1: The Initial Compromise

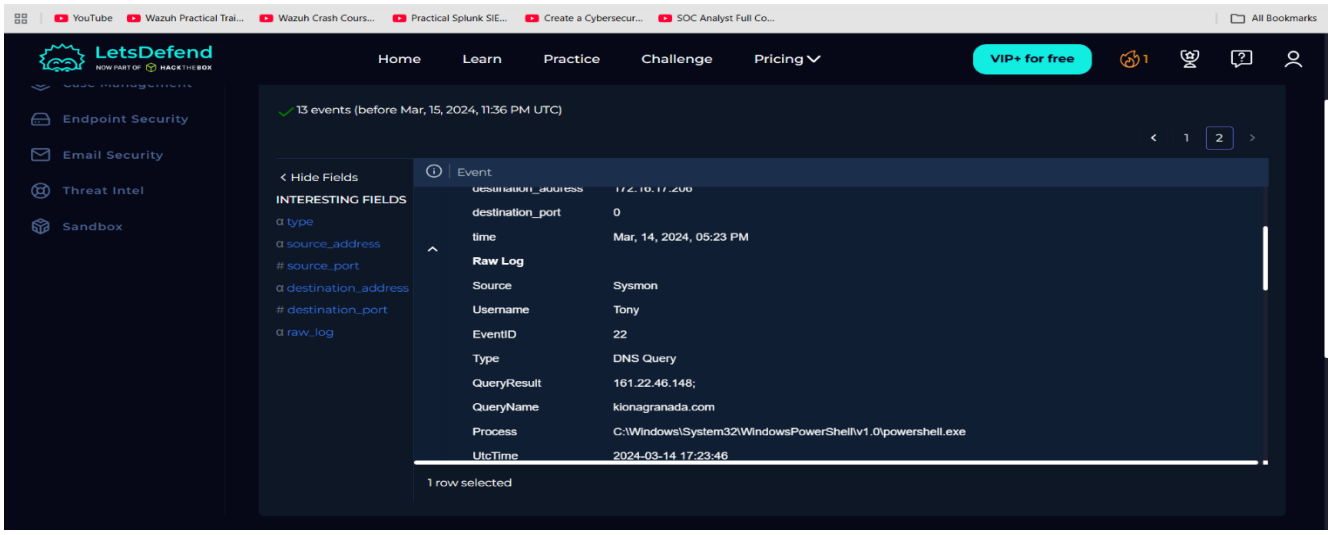


Initial access log shows the user LetsDefend using chrome.exe to download the malicious script payload_1.ps1 from a storage folder on the internet. The device action was allowed which means the infection was started.

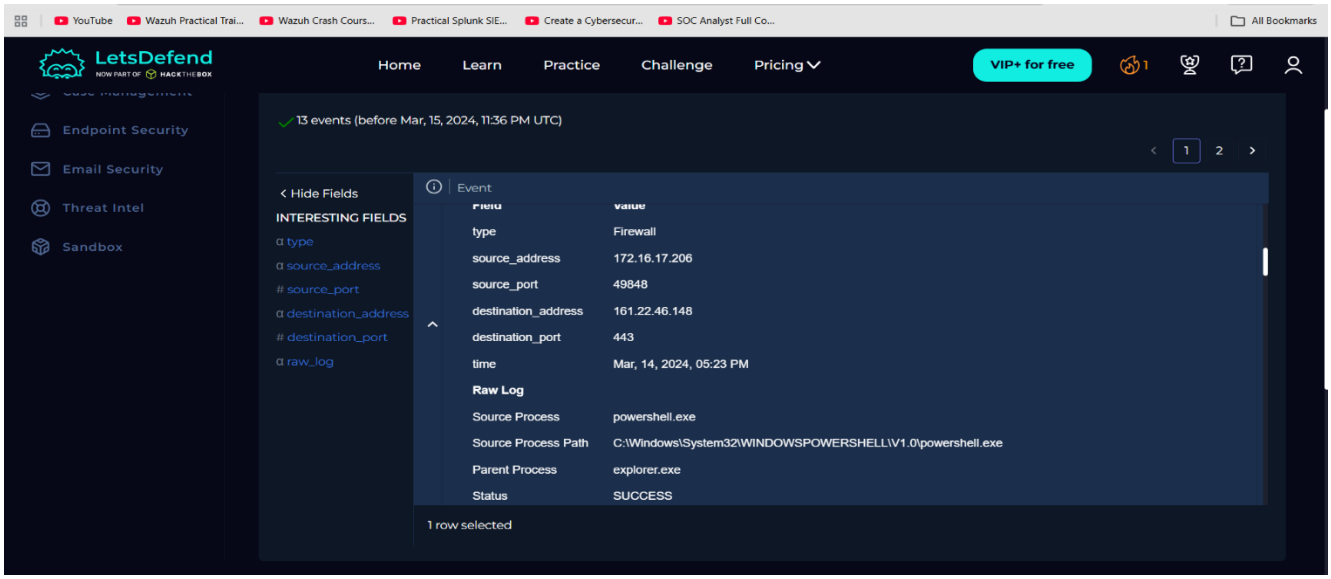


Windows event ID 1 captures the execution of the script that was downloaded payload_1.ps1. The log also shows the file payload_1.ps1 is linked to PowerShell process ID 4315 which confirms user ran the malware.

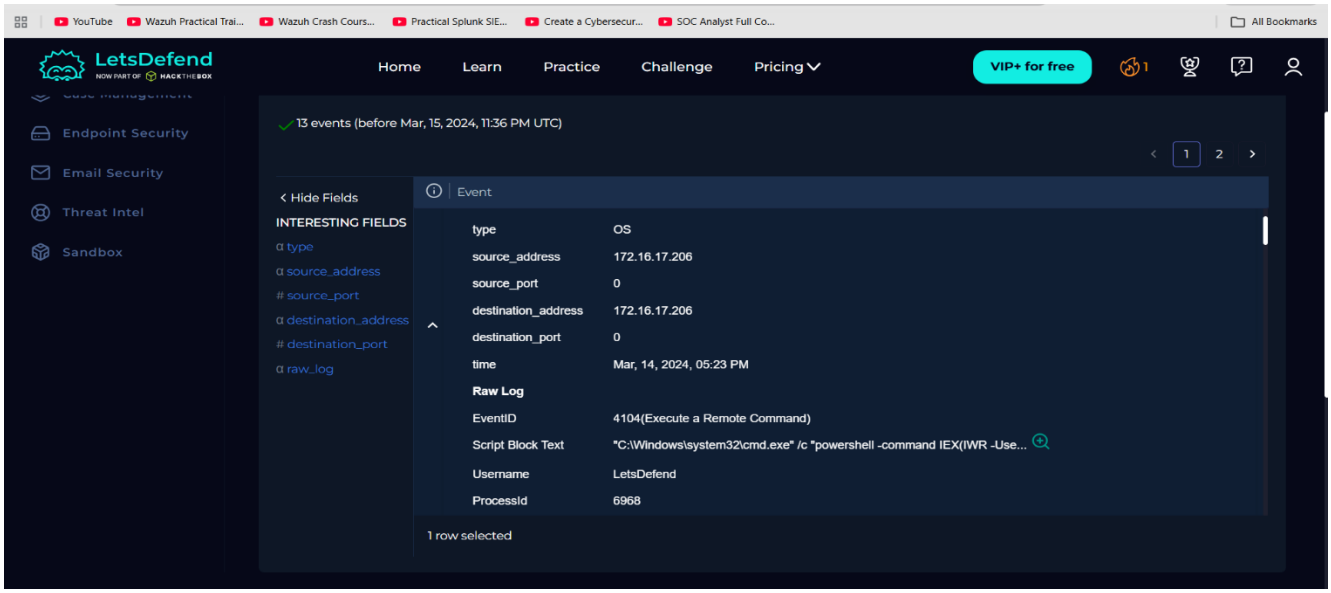
Phase 2: Command & Control



Sysmon Event ID 22(DNS Query) showing the script attempting to query domain kionagranada.com. This identifies attacker's command and control channel.

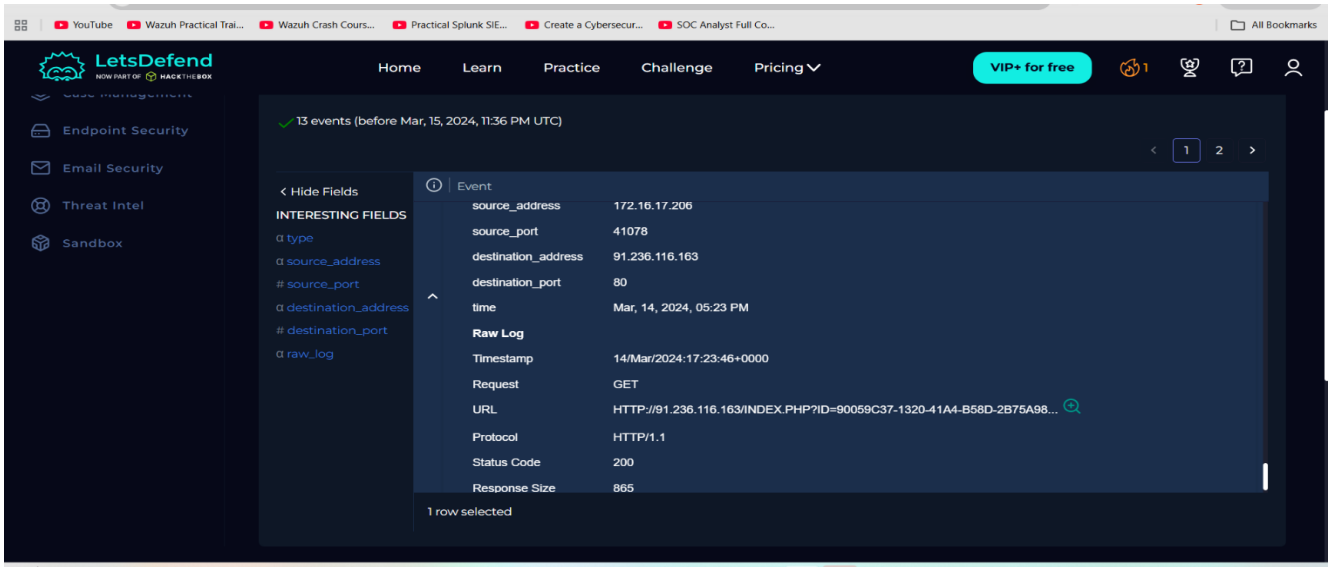


This log shows a successful outbound connection from the workstation to malicious IP 161.22.46.148 on port 443. This means the attacker's server was now talking to the workstation and ready to send more commands.

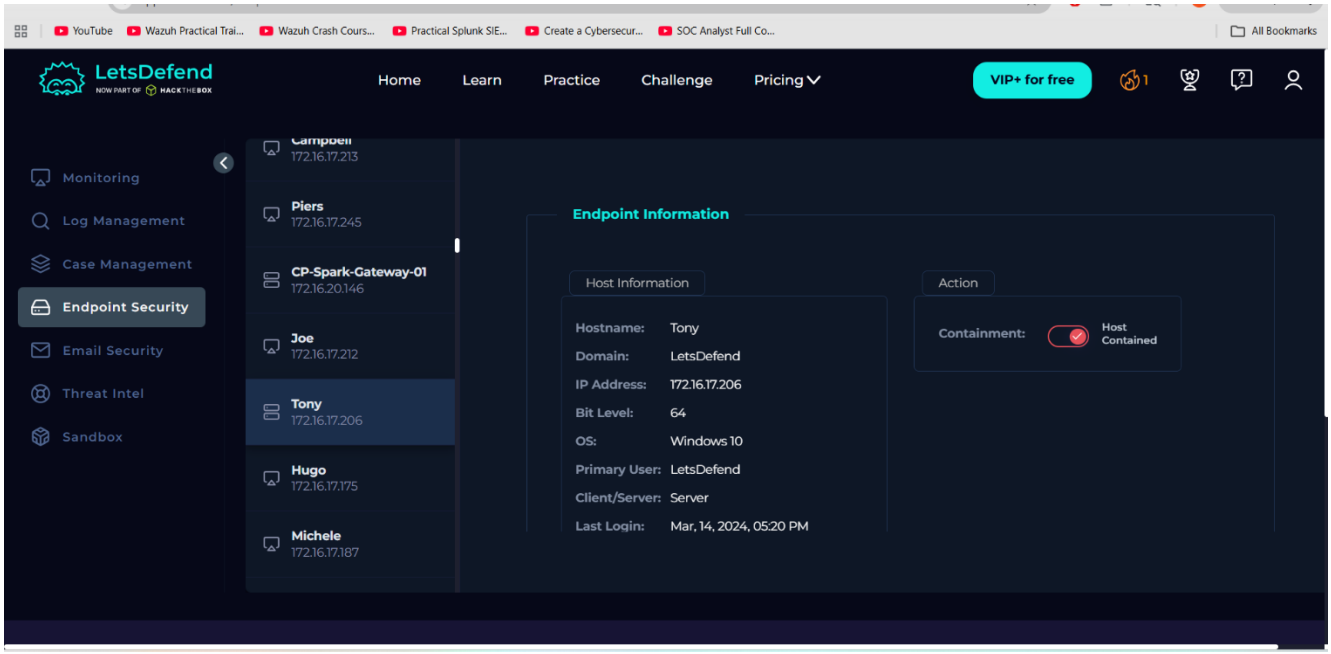


EventID 4104 shows remote command execution. The script block shows the command using IEX and IWR to download and execute the second stage payload sd2.ps1 into the memory.

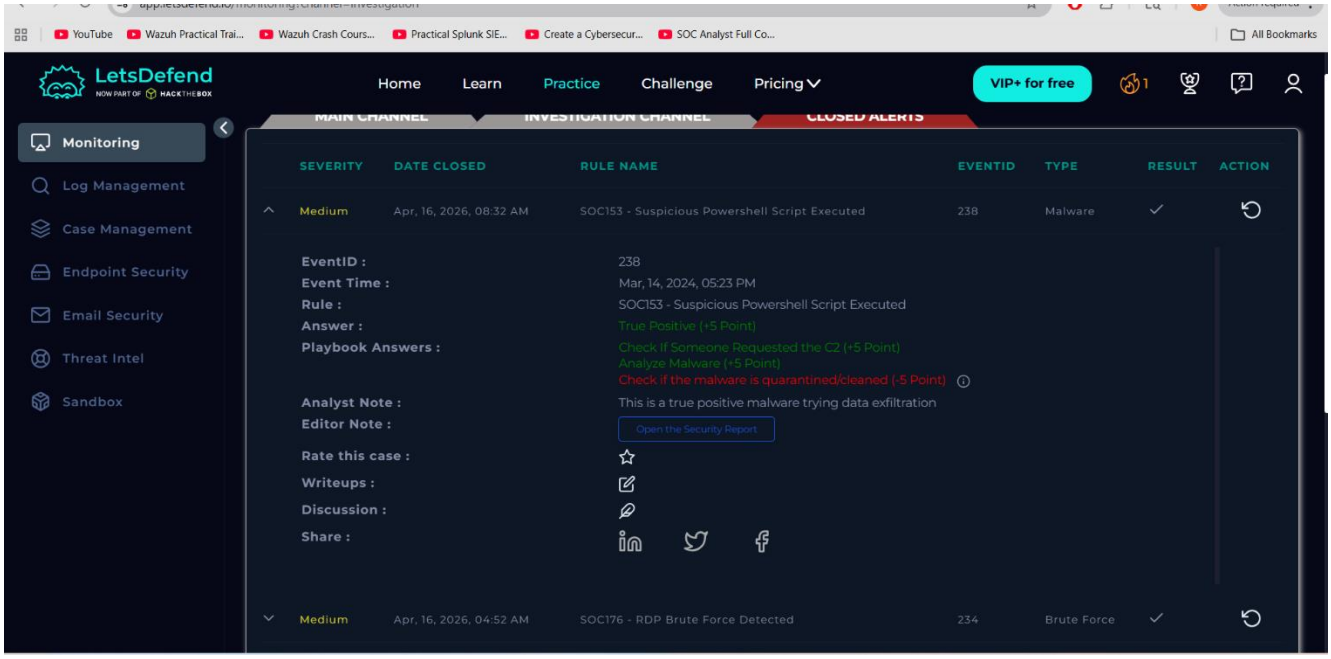
Phase 3: Impact and Containment



HTTP get request log showing active data exfiltration to IP 91.236.116.163. The status code 200 and the URL confirm the sensitive system information was successfully sent to the attacker's server.



After the detection, the host was isolated. This action terminated all C2 sessions and prevented further data theft.



The Summary of the findings.

6. Recommendations & Remediation

Short-Term actions:

1. Block the Domain and IPs: Block the domain kionagranada.com and the IPs it used 161.22.46.148, 91.236.116.163 and 66.81.199.52
2. Credential reset: Change the password for the username “Letsdefend” and any other accounts that were logged into the machine during the infection.
3. Scanning other workstations: Scan other workstations in the network to find the file payload_1.ps1 to ensure the virus hasn’t spread.
4. Terminate the process: Identify and Kill the active PID 4315.

Long-Term actions:

1. DNS Filtering: To block low reputed sites
2. Unused services: If any service like PowerShell is not needed for daily use, then disable it
3. User Awareness Training: Conduct training for all the company employees on malicious download. Teach employees to keep a check on what they are downloading and if it is a legit site or a fake one.